Samarvir Garg

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Education

University of Toronto Torornto, ON
Bachelor of Applied Science and Engineering - Computer Engineering *Sep. 2025 – Present*

White Oaks Secondary School Oakville, ON
Founder of Math Support Club, Math and Physics Tutor, Executive of Hindu Association *Sep. 2023 – June 2025*

Technical Skills

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS
Frameworks: Spring Boot
Developer Tools: Git, VS Code, Visual Studio, IntelliJ, Eclipse
Libraries: Pygame, Becker

Experience

Software Trainee Oct. 2024 – May 2025
Master Software Solutions *Mohali, IND*

- Contributed in development of route optimization algo for delivery management SaaS platform www.trakop.com
- Training in the programming language Java Spring and PostgreSQL database
- Worked on an analytics dashboard on historical datasets for sales, customers, orders, payments and inventory

Research Assistant (Volunteer) Nov. 2024 – Feb. 2025
University of Toronto *Toronto, ON*

- Conducted research under Prof. Clement C. Zai on ”Pharmacogenetic Reporting and Data Visualization in Mental Healthcare”
- Reviewed healthcare data reporting systems, including clinical dashboards and decision-support tools.
- Authored a technical report proposing scalable, data-driven visualization design recommendations.

Member, Driverless – Deep Perception Subteam Jan 2026 – Present
University of Toronto Formula Racing *Toronto, ON*

- Working on ML-based perception for cone detection, classification, and localization using camera image streams.
- Contributing to data collection and manual labeling to improve model generalization.
- Evaluating models using IoU, mAP, and inference latency under real-time NVIDIA Jetson constraints.

Team Lead – APS111: Engineering Strategies & Practice II Sep. 2025 – Dec. 2025
University of Toronto *Toronto, ON*

- Led a multidisciplinary team on a client-based engineering design project for **High Park**.
- Developed key design deliverables including **Technical Criteria (TC)**, **Problem Requirements (PR)**, and **Concept Design Specifications (CDS)**.
- Coordinated team workflow, timelines, and final design reports and presentations.

Projects

Emergency Route Optimization System | *Java, Becker Robots, SQLite, REST API* Jan. 2026

- Built a backend routing engine implementing priority-based, maps-based path planning and multi-agent coordination for ambulance dispatch
- Developed a lightweight local server (REST-style API) to handle emergency inputs, ambulance status updates, and routing decisions between system components
- Integrated a SQLite database for persistence of emergency data and simulation logs, and validated logic through a Java-based simulation frontend using Becker Robots; placed 3rd overall at UTEK 2026

FoodChain — FBLA Project | *Java, Spring Boot, MySQL, MVC Architecture* Feb. 2024

- Designed and developed a full-stack backend system using Java Spring Boot following a 3-layer (Controller–Service–Repository) architecture
- Implemented RESTful APIs for managing restaurants, private chefs, and user search and filter functionality, backed by a MySQL relational database
- Built and tested the application on a local server, applying CRUD operations, dependency injection, and modular backend design principles